清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

أرامكو السعودية
saudi aramco

Assessment of Iron and Steel Technologies from A Global Perspective

Tsinghua University & Saudi Aramco

aramco

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

Presenter: Yunchuan Liu, PhD student

December 1, 2025

Project Timeline

清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July
Research Proposal & preparations																								
Overview of global steel trade & technology assessment																								
Data requirement & database investigation (Base on case India)																								
Model upgrade (Base on case India)																								
Case study: India																								
Case study: USA & Saudi Arabia																								
Case study: Australia & Japan																								
Case study: Germany & Russia																								
Case study: China-Australia steel trade route																								
Final report																								

Note1: Models and data requirements lead to each other's updates, so we often perform these two processes at the same time.

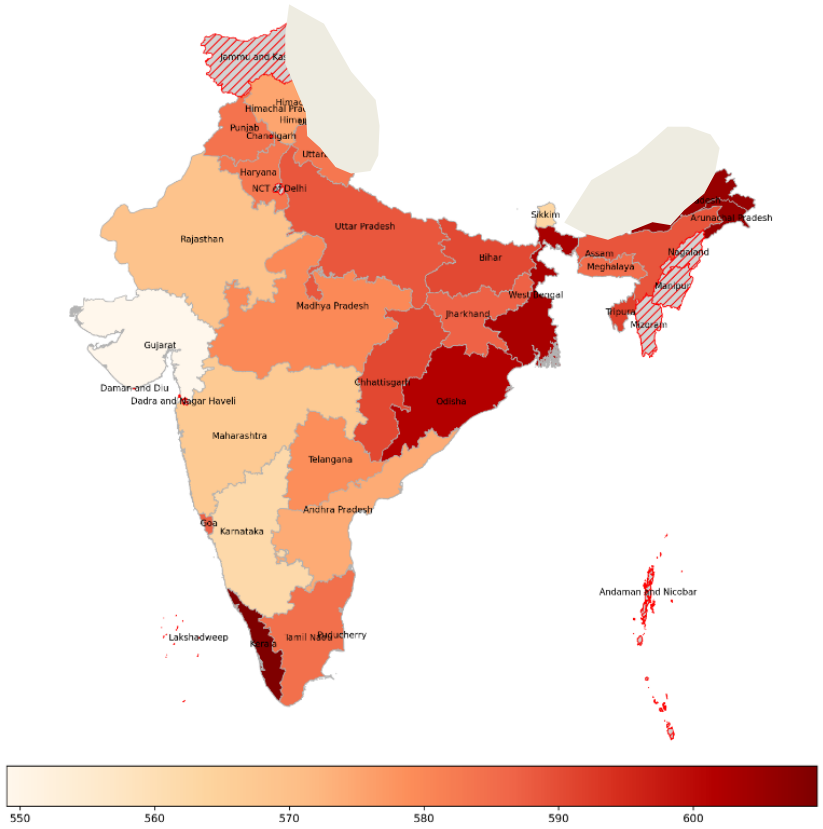
Note2: As models and data are continuously updated, case studies will be needed to assess whether they're effective - so we use the Indian case as an example.

Research update

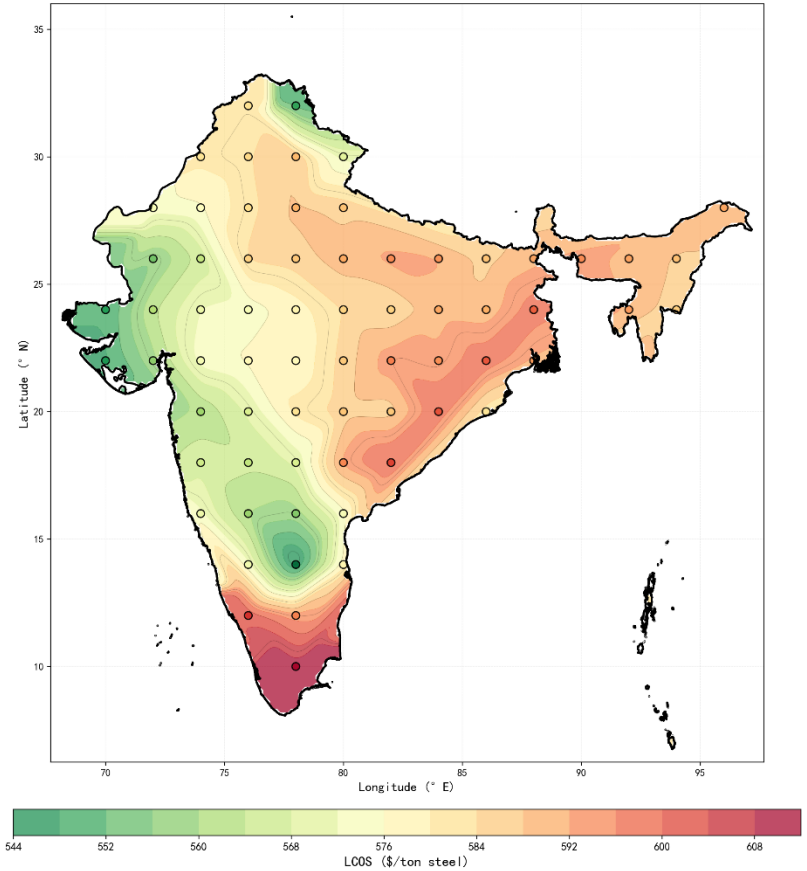
- India scenario display optimization
- Model development
  - Hydro-based H2-DRI-EAF
  - H2-DRI-ESF-BOF/EAF
- Case Study: Long-Term Transformation of China's Steel Industry (ESF considered)

Previous results: Indian case

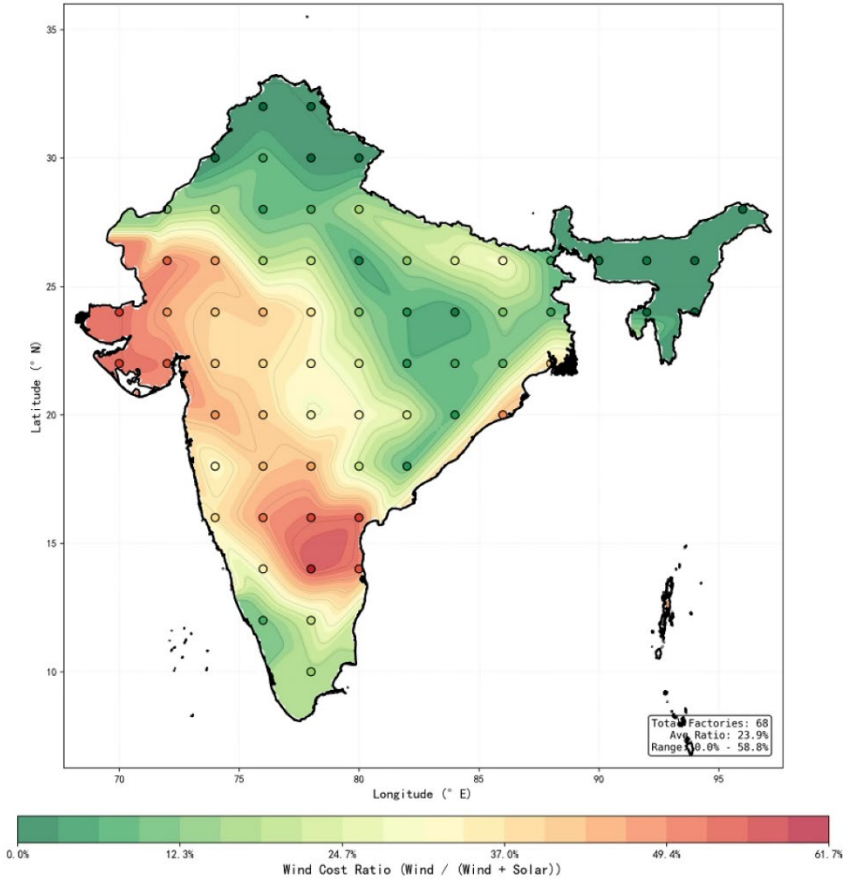
- The western India and south-central India have both higher wind energy content and lower LCOS.
- The LCOS in India exhibits significant **regional differences**, and we found that **the complementarity of wind and solar power** is a key factor contributing to cost differences.



LCOS for the H_2-DRI-EAF route in Indian states (USD/tcs, hatched areas excluded)



LCOS heatmap of the H_2-DRI-EAF route in major regions of India (USD/tcs)



Heat map of **wind power cost share** for H_2-DRI-EAF routes in major regions of India

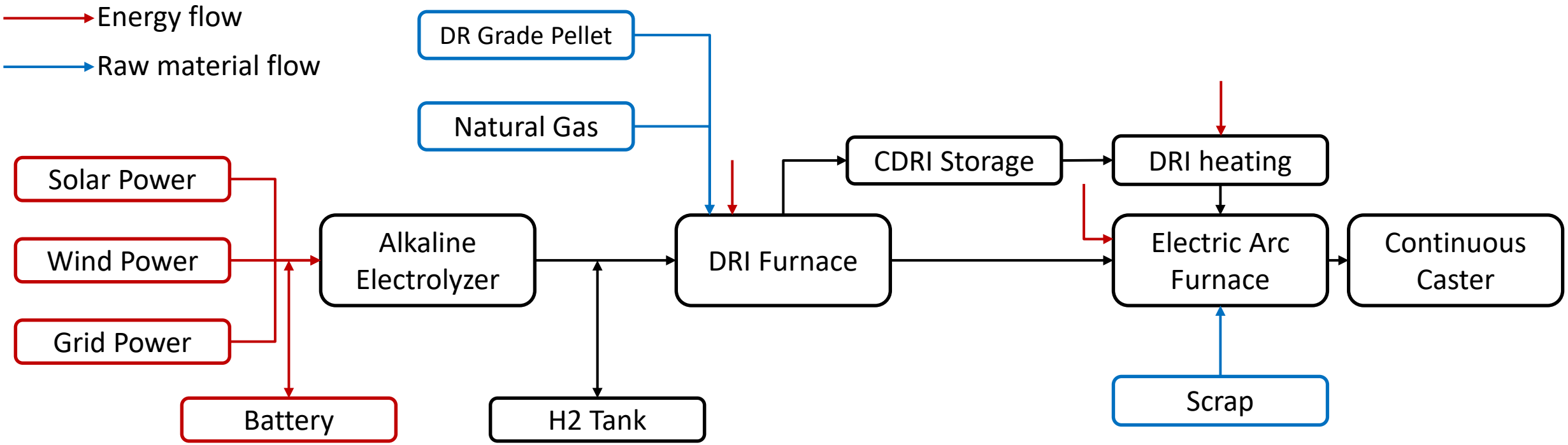
Methods: Current model

H2-DRI-EAF production scheduling model with flexibility

Hourly resolution system optimization model, **Minimizing LCOS**

Consider the **production, storage and consumption** of **electricity, hydrogen, iron and steel**

Flexibility modules: battery, H2 tank, CDRI storage

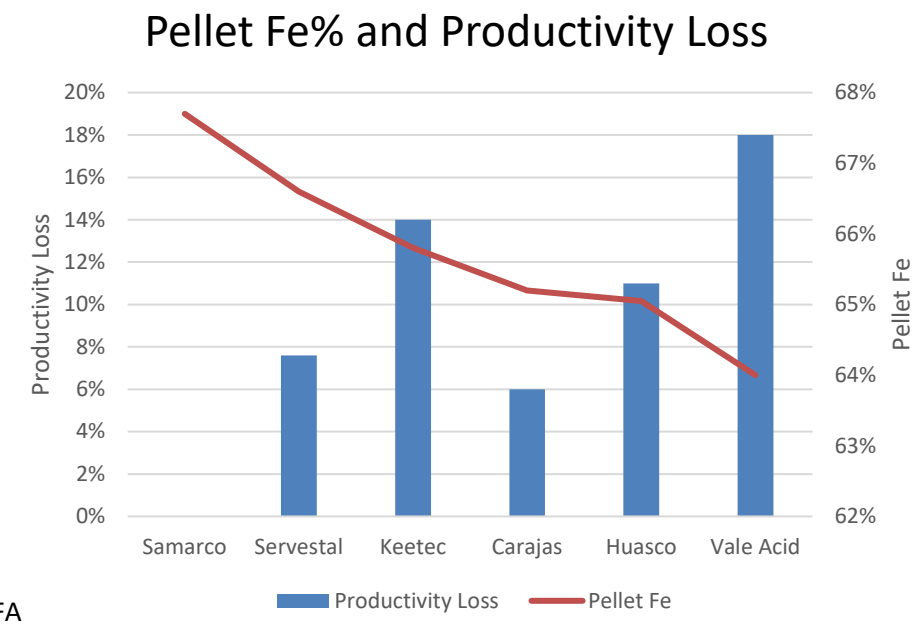


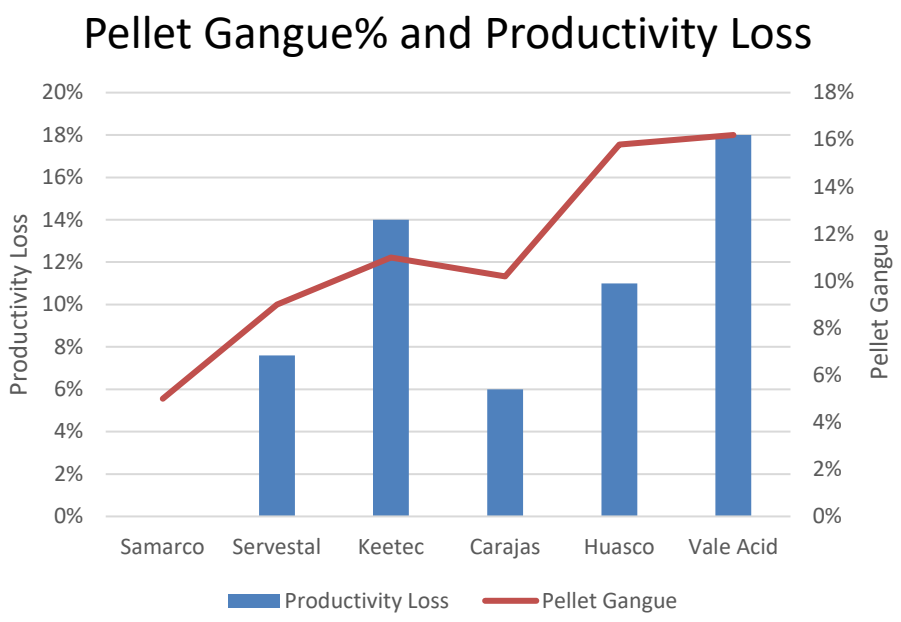
We will continue to update this model in subsequent research, and update the results in the Indian Case.

Question: What if H2-DRI-EAF is highly used?

Challenges: Traditional H2-DRI-EAF Route

- The traditional H2-DRI route relies on DR-grade pellets characterized by high iron content and low gangue/impurity levels.
- Using lower-grade BF-grade ore leads to increased electricity consumption and higher slag generation within the EAF.
- Even a slight increase in impurities can result in iron losses of up to 15%, which is almost unaffordable.





Source: IEEFA

Question: What if H2-DRI-EAF is highly used?

- Currently, only about **4%** of the global iron ore supply meets DR-grade standards.
- Against the backdrop of the global scale-up of H2-DRI, the limitations of the traditional H2-DRI-EAF route are being sharply magnified.
- To advance the decarbonization of the steel industry, it is imperative to develop new technology combinations, that enable the use of abundant low-grade iron ore for green steel production.

DR-grade Iron Oxide Chemical Quality Limits

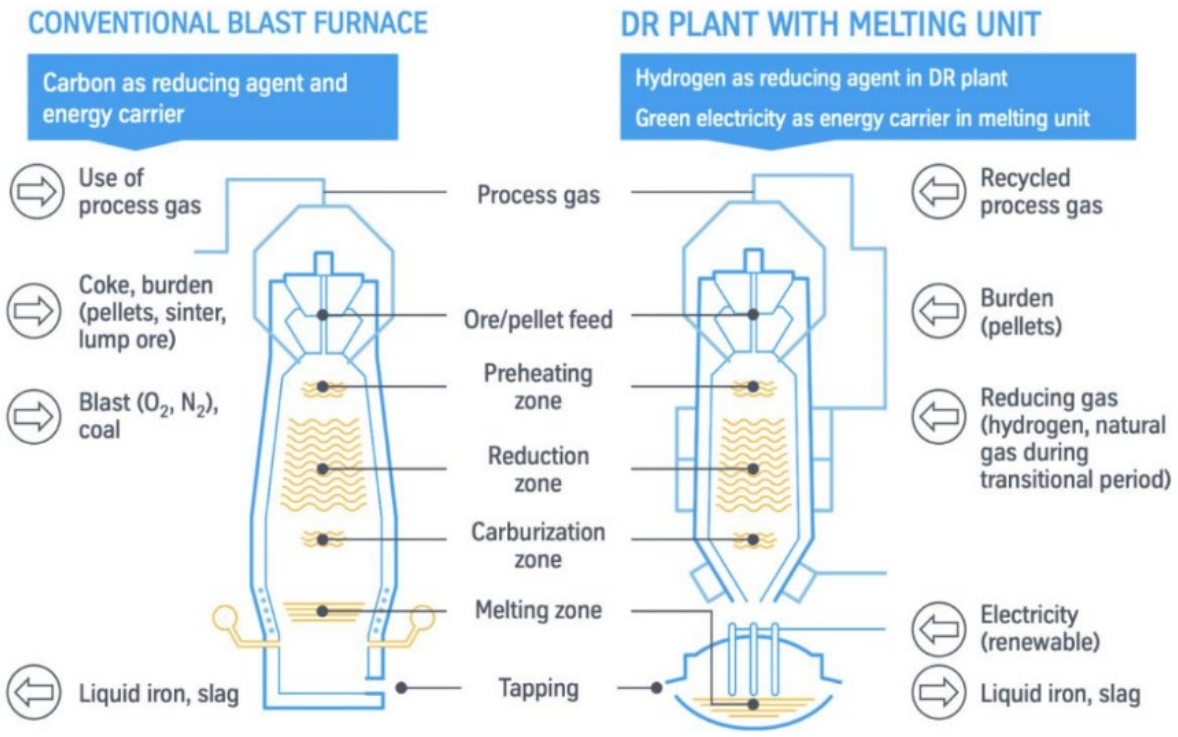
Sources	Practical Limits (%)	Preferred Limits (%)
Iron (Fe)	≥66.0	≥67.0
Silica (SiO_2) & Alumina (Al_2O_3)	≤3.5	≤2.0
Calcium oxides (CaO)	≤2.5	
Magnesium oxide (MgO)	≤1.0	
Phosphorous pentoxide (P_2O_5)	≤0.03	≤0.015
Sulphur (S)	≤0.025	≤0.015
Copper (Cu)	≤0.03	≤0.01
Titanium dioxide (TiO_2)	≤0.35	

Source: Midrex

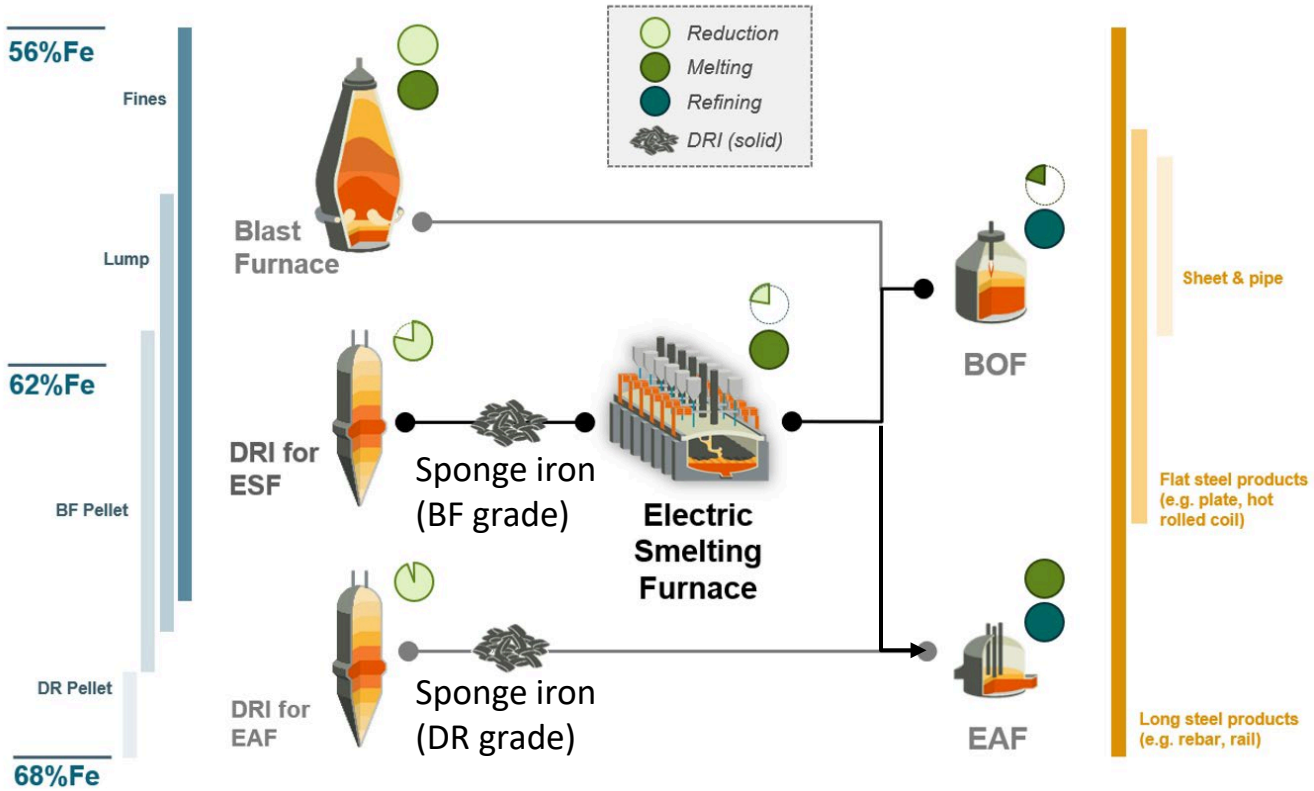
Question: What if H2-DRI-EAF is highly used?

H2-DRI-ESF-EAF/BOF route

- Following DRI production, the solid DRI is charged into the ESF, where it is melted and the slag is separated.
- The resulting liquid iron (1400-1500°C) can be directly charged into **BOF** or **EAF**.
- This route accommodates **BF-grade low-quality iron ore**, yielding steel with lower impurity levels.
- The ESF process involves maintaining continuous high temperatures, requiring **stable production rates**.



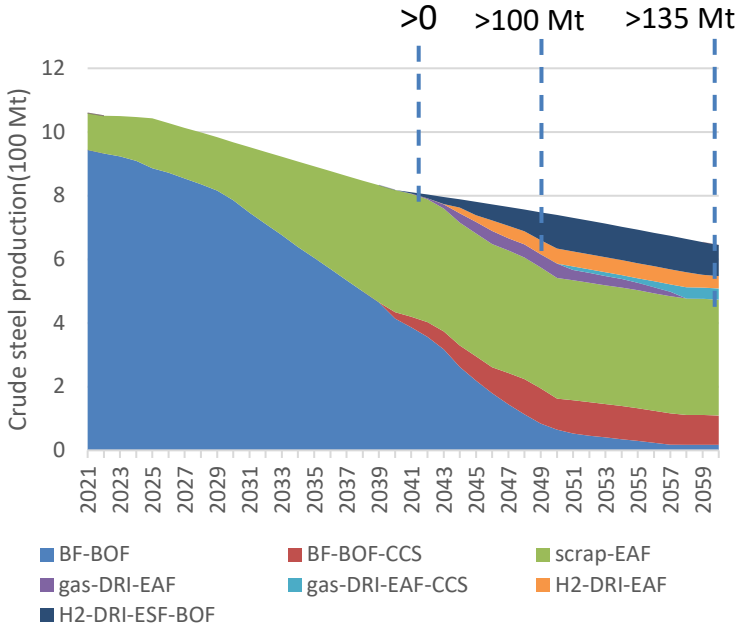
Source: Thyssenkrupp, BHP Billiton



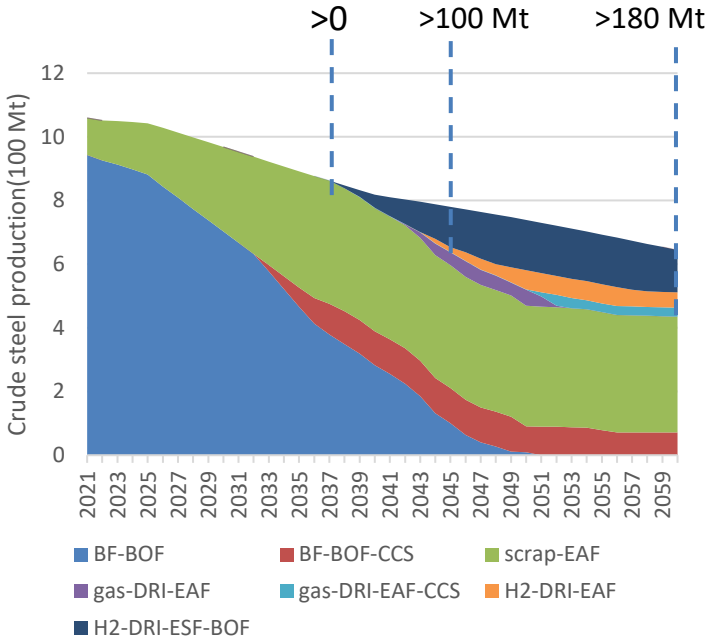
Case study: China

- This study demonstrates the technological structure for two scenarios, where the steel industry will be largely decarbonized by 2060 and 2050, after incorporating the ESF-BOF route.
- After considering the DRI-ESF-BOF route, it can be found that China's DRI-ESF-BOF production share is higher than that of DRI-EAF.
- The H2-DRI-ESF route can accept BOF access, making it cheaper to convert existing BF-BOF to H2-DRI-ESF-BOF.
- This study **partially reveals the importance of the H2-DRI-ESF-BOF/EAF route**.

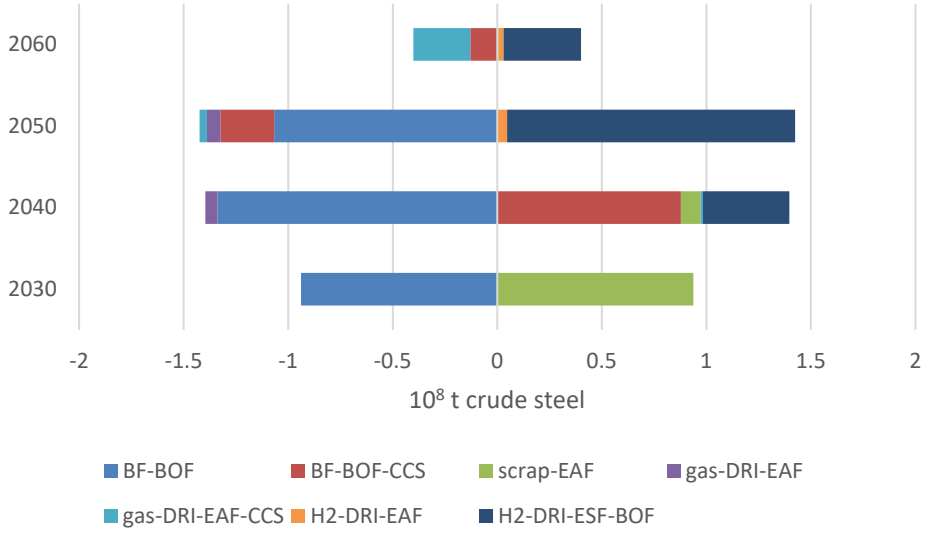
Technology mix– Scenario 2060



Technology mix– Scenario 2050

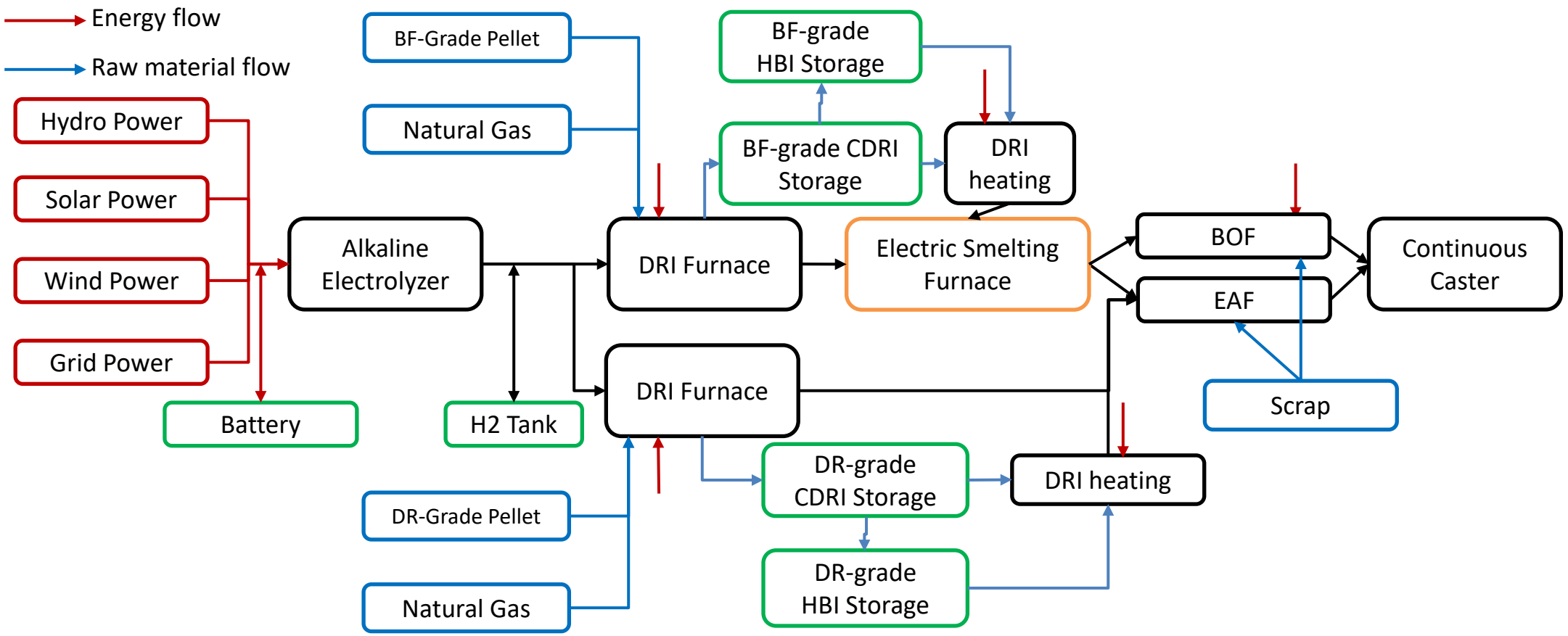


Changes in the 2050 Scenario compared with the 2060 Scenario



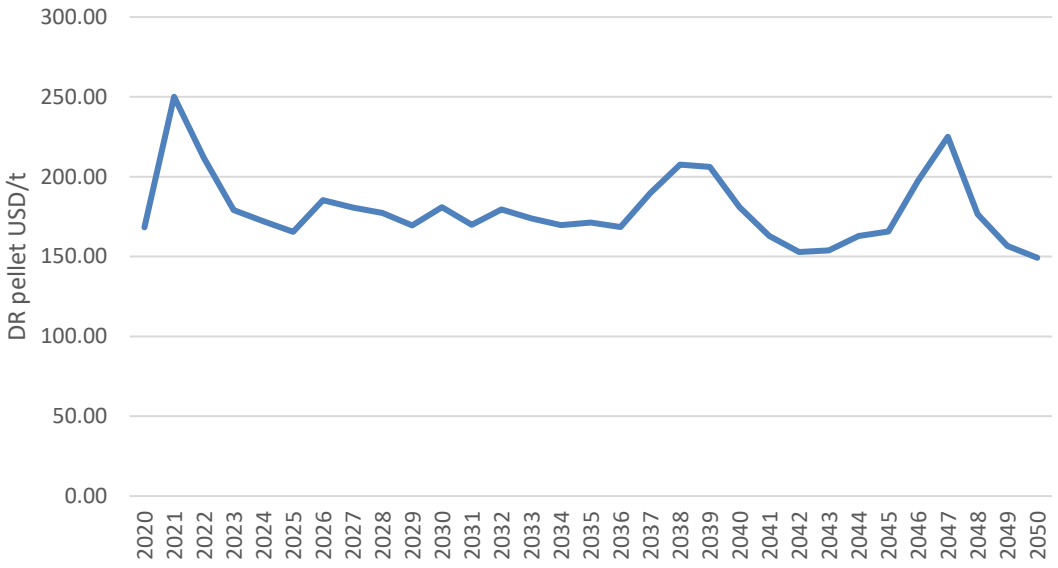
Methods: Upnext model

- The new model establishes **relatively independent pathways** for both BF-grade and DR-grade materials.
- By applying appropriate constraints, the model can simulate the competition and synergy among multiple routes: **BF-grade DRI-ESF-BOF**, **BF-grade DRI-ESF-EAF**, and **DR-grade DRI-EAF**.
- Hydropower is considered a resource form of equal importance to solar and wind power.

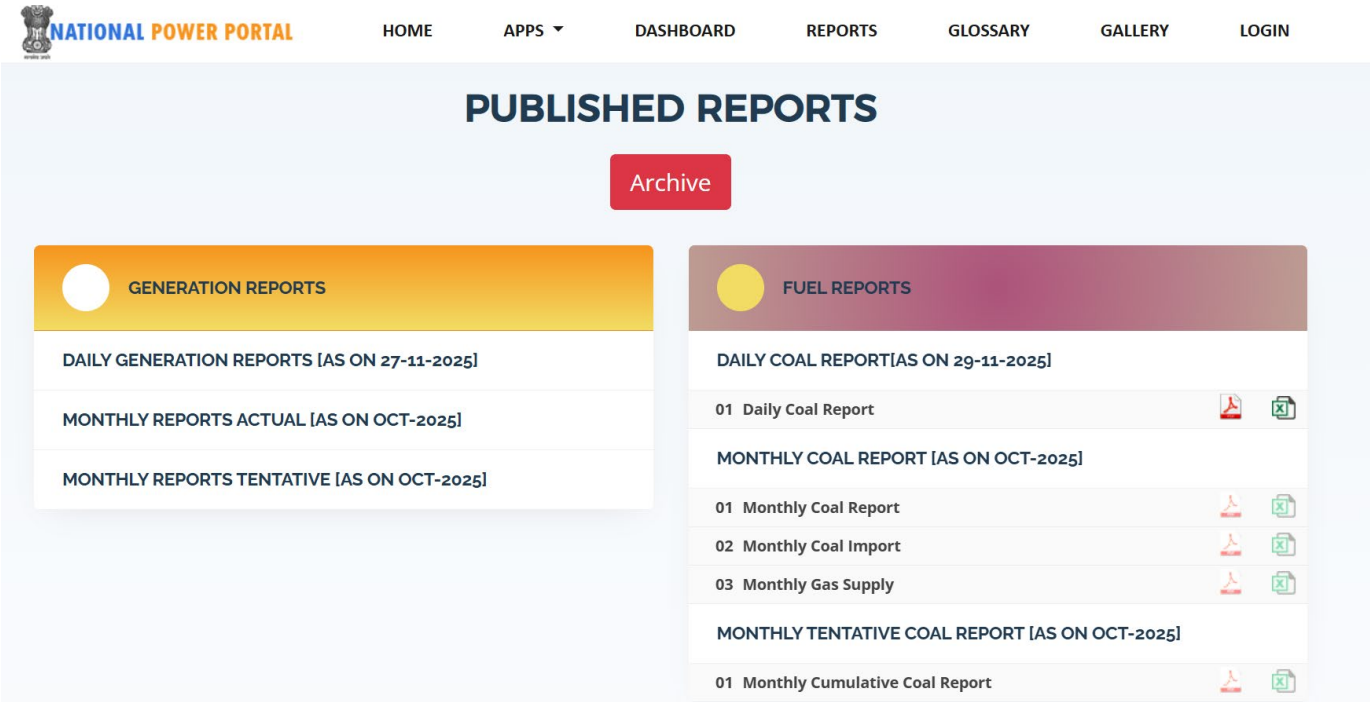


Data collection

- **BF-grade Pellet Price:** proprietary, Steelonthenet
- **DR-grade Pellet Price:** Collecting (we may use forecasted values based on various green steel development scenarios, or proprietary data from Steelonthenet)
- **ESF CAPEX/OPEX**: similar as EAF
- **Hydropower price**: Typical prices during the dry season and the wet season, electricity prices at other times are simulated.

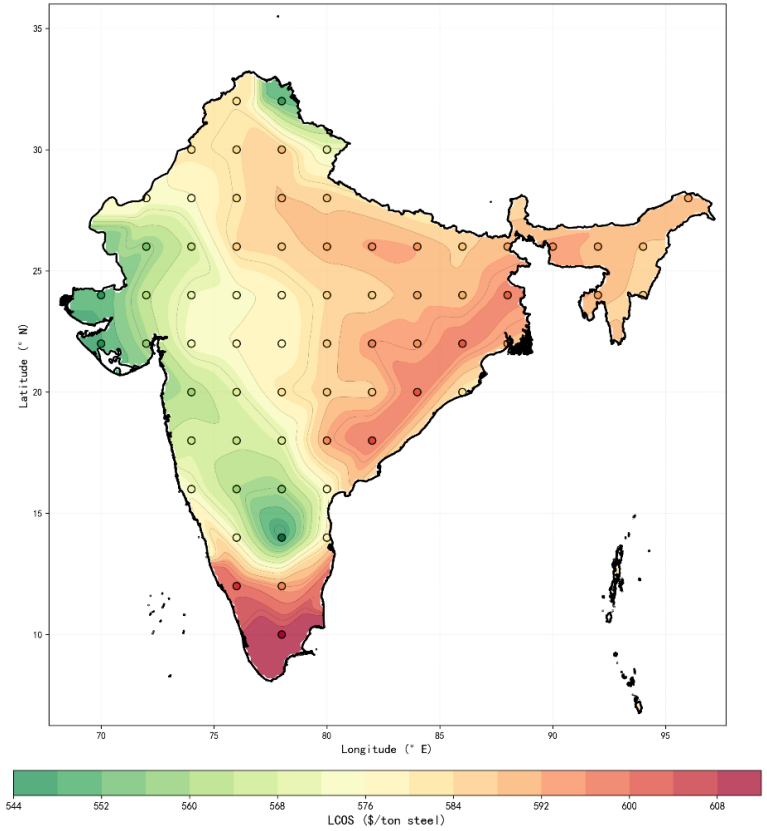


DR pellet price prediction on Steelonthenet, USD/t

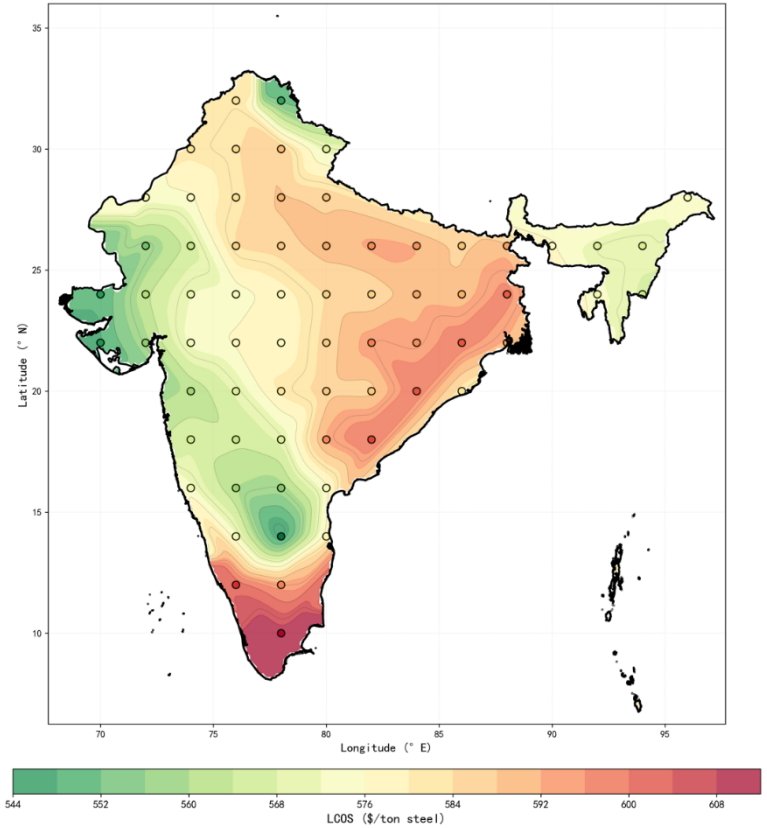


With hydropower considered…

- In northeastern India, the LCOS has decreased significantly.
- This is because hydropower supply is insufficient during the dry season, making large-scale substitution impossible.



LCOS heatmap of the H_2-DRI-EAF route in major regions of India **without** Hydropower considered(USD/tcs)



LCOS heatmap of the H_2-DRI-EAF route in major regions of India **with** Hydropower considered(USD/tcs)

Next steps

- Applying H2-DRI-ESF-BOF/EAF in the model code
- Complete data collection
- Case study: Long-Term Transformation of India's Steel Industry

Thank You For Your Attention!